Ke Li

Personal Website: https://kelichloe.github.io

Research Interests

My research lies at the intersection of AI/ML, optimization, and decision making under uncertainty. I pursue two complementary streams:

- **Decision-Aware Learning**: developing machine learning and data-driven optimization methods that align learning objectives with downstream managerial decisions.
- **Human–AI Collaboration / AI for Science**: studying how AI can augment researchers' workflows, particularly in scientific reasoning and theory building.

Education

INSEAD France
PhD Candidate in Decision Sciences 09/2023 - Ongoing
Advisor: Spyros Zoumpoulis

The Chinese University of Hong Kong, Shenzhen China
B.Eng (Honors), Computer Science and Engineering 09/2018 - 06/2022

Honors & Awards

Academia and Research

– OpenAI Researcher Access Program ($1,000 API credits)	2026
– *Stars of Tomorrow* Award for Outstanding Intern, Microsoft Research Asia	2022
– ExploreCSR Computing Research Award, Google Research & UTRGV [link]	2021
– China National Encouragement Scholarship, Ministry of Education of China	2019
– Dean's List, CUHK-Shenzhen	2019–2020
– Undergraduate Admission Fellowship (CNY 90,000/year), CUHK-Shenzhen	2018–2022

Teaching

– Teaching Excellence Award in PhD Tutorials, INSEAD	2025
– Undergraduate Student Teaching Fellow, CUHK-Shenzhen	2020

Working Papers

[1] **Decision-Aware Customer Segmentation**. *Coauthors*: Sandeep Chandukala (SMU), Ernst Osinga (SMU), and Spyros Zoumpoulis (INSEAD).
Presentations:
- **2026:** Best Paper Award, Singapore Rising Scholars Conference (SMU); INFORMS Annual Meeting (San Francisco); M&SOM Conference (Harvard); ISMS Conference (Lisbon); Theory + Practice in Marketing (Esade).

[2] **Humans versus and with AI for Theory Building**. *Coauthors*: Philip Parker (INSEAD), Phanish Puranam (INSEAD), Eric Uhlmann (INSEAD), and Spyros Zoumpoulis (INSEAD).
Presentations:
- **2025:** INFORMS Annual Meeting (Atlanta); AI Plus Management Doctoral Consortium (UCL).

Publications

See my google scholar page.

[1] Zhou, B., Li, K., Jiang, J., & Lu, Z. Learning from Visual Observation via Offline Pretrained State-to-Go Transformer. *Advances in Neural Information Processing Systems* (**NeurIPS**, Top-tier AI/CS/ML conference), 2023.

[2] Dong, J., Li, K., Li, S., & Wang, B. Combinatorial bandits under strategic manipulations. In *the Fifteenth ACM International Conference on Web Search and Data Mining* (**WSDM**, Top-tier data mining conference), 2022.

[3] Liu, Y., Li, K., Huang, Z., Li, B., Wang, G., & Cai, W. EduChain: a blockchain-based education data management system. In *Blockchain Technology and Application: Third CCF China Blockchain Conference*, 2021.

Industry Experiences

Beijing Academy of Artificial Intelligence Beijing
Reinforcement Learning Researcher (Full-time) 12/2022 - 07/2023

- Designed and trained intelligent agents capable of learning directly from high-dimensional visual observations in complex game environments such as Minecraft.
- Developed novel architectures combining convolutional encoders with deep reinforcement learning algorithms (e.g., PPO, DQN)
- Led extensive experiments to validate model performance, and collaborated closely with researchers from Peking University. Our research paper (website) is accepted by NeurIPS 2023.

Microsoft Research Asia Beijing
Research Intern, Award of Excellent Intern, Innovation Engineering Group 01/2022 - 06/2022

- Investigated model compression techniques for Transformer/BERT models, such as pruning, quantization, and distillation.
- Implemented compression pipelines and reproduced results from recent research papers.
- Optimized Transformer models for deployment by reducing parameters and memory footprint.
- Collaborated with researchers to refine experimental settings and improve reproducibility.

SenseTime Shenzhen
Reinforcement Learning Intern, OpenDI Lab 08/2021 - 12/2021

ByteDance Technology Beijing
Machine Learning Intern, Data AI Lab 12/2020 - 05/2021

Contributions to Open-Source Community

- **Microsoft Research Asia** — Contributed to Microsoft's AutoML toolkit NNI (14,300+ GitHub stars), including transformer model-compression modules now integrated into both internal and external AutoML pipelines.
- **SenseTime (OpenDI Lab)** — Contributed 1,600+ lines of code to the DI-engine reinforcement-learning framework (3,600+ GitHub stars), improving algorithmic stability and large-scale training utilities.

Teaching Experience

Teaching Assistant, INSEAD
Winner of Teaching Excellence Award
- Foundations of AI for Managers, MBA Course 2025.05 - 2025.06
- Foundations of AI for Managers, MBA Course 2025.01 - 2025.02
- Probability and Statistics, PhD Course 2024.10 - 2024.12

Teaching Assistant, CUHK-Shenzhen
- MAT1010 Calculus I 2019.09 - 2019.12

Academic Services

Session Chair
- INFORMS Annual Meeting (co-chair with Professor Spyros Zoumpoulis) 2026
Session Title: Advances in Data-Driven Decisions
- INFORMS Annual Meeting (co-chair with Professor Spyros Zoumpoulis) 2025
Session Title: Estimation and Optimization in Data-Driven Decisions

Conference Reviewer
- NeurIPS (Top-tier conference in CS/ML/AI) 2024

Beyond Academia

Part-time model in Paris ([portfolio]).
Piano Grade 10.

Media

Student interview by CUHK-Shenzhen (EN version/CN version)

Patents

Personnel exemption aided decision making method and system based on historical big data. China Patent CN113673943A. [link]